

UNIVERSITY OF MACAU
FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER AND INFORMATION SCIENCE
2026-1-CISC3011 DIGITAL IMAGE PROCESSING

Syllabus
1st Semester 2026/2027
Part A – Course Outline

Course Description

Digital Image Processing (DIP) is a fundamental and important component in numerous real-world applications, such as business, industry, medical imaging, digital photography, computer games, robotics, etc. Its applicability enables effective and efficient communication through visual results, e.g., tele-presence and tele-communication for metaverse. This 3-credit, 14-week course prepares students with the fundamentals and various applications of DIP. Students will gain the ability of developing appropriate DIP techniques to solve real-world problems.

Course Type

Theoretical

Prerequisites

- Python programming
- Linear algebra
- Statistics and probability
- Multivariate calculus
- Algorithms
- Preferably access to GPUs (e.g., Public Lab in E11, [Google Colab](#)).

Recommended Background

- **Python.** All class assignments will be in Python. For those who aren't as familiar with Python, please follow this [tutorial](#). In the tutorial, go over all topics under "Python Tutorial", "Python NumPy". It will also be beneficial to go over all topics under "Machine Learning".
- **Familiar with Jupyter Notebook.** We will use Jupyter Notebook (<https://jupyter.org>) to implement algorithms, demonstrate experiment qualitative and quantitative results, and write up assignment report. There is an online lab of Jupyter (<https://jupyter.org/try-jupyter/lab>). Converting a Jupyter Notebook file to pdf script is trivial (<https://nbconvert.readthedocs.io/en/latest/usage.html>).
- **PyTorch.** Some coding assignments can be done in a more convenient way by using deep learning. [PyTorch](#) is an excellent python-based toolbox for machine learning and deep learning. For those who haven't used it before, please refer to its official [tutorial](#) and navigate specific topics therein such as "Image and Video".
- **Linear Algebra.** You should be comfortable taking derivatives and understanding matrix vector operations and notation. Go over the "Essence of linear algebra" playlists by "3Blue1Brown" at <https://www.youtube.com/c/3blue1brown/playlists>
- **Statistics and Probability.** You should be familiar with basics of probabilities, Gaussian distributions, mean, standard deviation, etc. Go over the "Probabilities of probabilities" playlist by "3Blue1Brown" at <https://www.youtube.com/c/3blue1brown/playlists>

Popular Textbooks and Reading Material

No textbooks are required; below are recommended. All learning materials are online available.

- Steven W. Smith, The Scientist and Engineer's Guide to Digital Signal Processing ([Online](#))
- R. C. Gonzalez, R. E. Woods, Digital Image Processing, 4th edition, Pearson, 2018
- William K. Pratt, Introduction to Digital Image Processing, CRC Press, 2013
- A. Kaehler, G. Bradski, Learning OpenCV 3, O'Reilly Media, 2017.
- IEEE Transactions on Image Processing (TIP)
- IEEE International Conference on Image Processing (ICIP)
- IEEE Computer Vision and Pattern Recognition (CVPR)
- [scikit-image: Image processing in Python](#)

Course Objectives

The objective of this course is to teach fundamental concepts, methods and problems related to Digital Image Processing. The course will also train students with respect to implementing classic digital image processing algorithms. Upon completion of the course, students will:

1. have a good understanding of fundamental concepts and classic algorithms in DIP,
2. know useful toolboxes for DIP,
3. be able to develop methods to solve real-world problems related to DIP,
4. build interests in other topics like computer vision, machine learning, and deep learning.

The learning outcomes will be assessed through homework, course projects, and exams.

Topics

1. Point operations/combining images/histograms
2. Image filtering, correlation, convolution
3. Edge, contour, corner
4. Morphological image processing
5. Color space
6. Image thresholding, segmentation
7. Scale-space image processing
8. Image matching, image registration
9. Applications: deblur, denoise, super-resolution, etc.

Class / Laboratory Schedule

Timetabled work in hours per week			No of teaching weeks	Total hours	Total credits	No of exam papers
Lecture	Tutorial	Practice				
1.5	0	1.5	14	52	3	2

Student Study Effort Required

Class:	
Lecture	21 hours
Tutorial	0 hours
Other study Effort:	
Self-study	21 hours
Course project	21 hours
Total student study effort	42 hours

Student Assessment

Final assessment will be based on

- Eight homework assignments: $72\% = 8 \times 9\%$
- Two exams (mid-term and final): $28\% = 2 \times 14\%$
- A bonus project: +10 points additive to the final exam, which is capped at 100.

Course Outline

Weeks	Topics	Homework / exam / project
1	Introduction, logistics	Homework 1: Basic python
2	Point operations, combining Images	Homework 2: Brightness and blending
3	Histograms	Homework 3: Histogram equalization
4	Filtering, convolution	Homework 4: Filter, blur, denoising
5	Edges	Homework 5: Edge, sharpening
6	Morphological processing	Homework 6: Morphological operation
7	Lines and corners	Mid-term exam
8	Scale space, compression	Homework 7: Image pyramid
10	Spectral methods, DFT, 2D Fourier	Homework 8: Spectral methods
11	Template matching, feature-based matching	Bonus project: Matching
12	Color science, color spaces	
13	Image restoration	
14	deep learning for digital image processing	Final exam

Student Disabilities Support Service:

The University of Macau is committed to providing an equal opportunity in education to persons with disabilities. If you are a student with a physical, visual, hearing, speech, learning or psychological impairment(s) which substantially limit your learning and/or activities of daily living, you are encouraged to communicate with your instructors about your impairment(s) and the accommodations you need in your studies. You are also encouraged to contact the Student Disability Support Service of the Student Counselling and Development Section (SCD), which provides appropriate resources and accommodations to allow each student with a disability to have an equal opportunity in education, university life activities and services at the University of Macau. To learn more about the service, please contact SCD at scd.disability@umac.mo, or 8397 4901 or visit the following website: <https://sao.um.edu.mo/>

Instructor

Shu Kong, Assistant Professor

<https://aimerykong.github.io>

Persons Who Prepared This Description

Shu Kong, August 10th, 2026

Part B General Course Information and Policies

1st Semester 2026/2027

Instructor: Prof. Shu Kong

Office: E11-4025

Office hour: Monday at 11:00-15:00, or by appointment

Email: skong@um.edu.mo

Time/Venue

Lecture	Tuesday 10:00-11:15am	E21 G016
Lab (with lecture)	Friday 10:00-11:15am	E11 1046

Grading Distribution

Percent. Grade	Final Grade	Percent. Grade	Final Grade	Percent. Grade	Final Grade
100 – 93	A	77 – 73	B-	57 – 53	D+
92 – 88	A-	72 – 68	C+	52 – 50	D
87 – 83	B+	67 – 63	C	below 50	F
82 – 78	B	62 – 58	C-		

Homework Policy

The completion and correction of homework are helpful for learning. As a result,

- There are five homework assignments.
- Homework is due one week after assignment unless otherwise noted.
- Late submission is subject to mark penalty
 - Within 3 days: -10%
 - 4-7 days: -30%
 - Over 7 days: -100%
- Possible revision of grades may be discussed with the instructor.
- The course grade will be based on the weighted average of the homework, midterm/final exam, and the course project.

Other Important Notes

- Check the ummoodle web pages for announcements, and lectures.
- Cheating is absolutely prohibited by the university.
- UM students can use ChatGPT or other generative-AI systems to **enhance** their learning and “must acknowledge the use of GenAI in all assignment submissions (written, spoken or otherwise) and indicate how GenAI has been used.” See details in FAQs for Students [updated 20 November 2025]: Using Generative AI Tools in Graded Assignments (<https://ctle.um.edu.mo/resource/faqs-for-students-using-generative-ai-tools-in-graded-assignments/>).